

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 9

INSOLATION

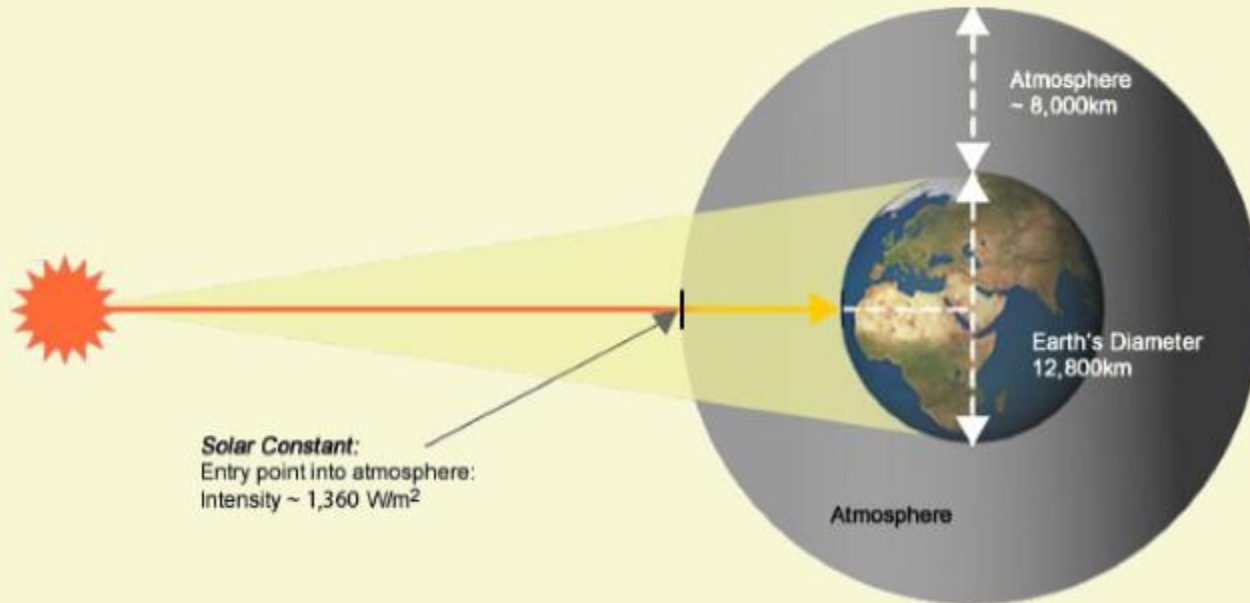
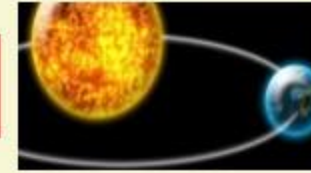
FACTORS CONTROLLING INSOLATION

CLIMATOLOGY MODULES

Atmosphere	▪ Composition and structure; weather and climate;
Insolation	▪ Factors controlling insolation; heat budget of the earth — heating and cooling of atmosphere (conduction, convection, terrestrial radiation, advection);
Temperature	▪ Factors controlling temperature; distribution of temperature — horizontal and vertical; inversion of temperature;
Pressure	▪ Pressure belts; winds-planetary, seasonal and local; air masses and fronts; tropical and extra-tropical cyclones;
Precipitation	▪ Evaporation; condensation — dew, frost, fog, mist and cloud; rainfall — types and world distribution;
World Climate and Climatic Changes	▪ Different types of Climate



INSOLATION

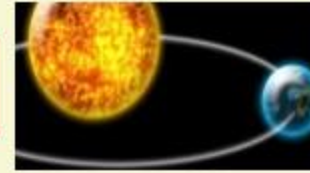


1 The energy received by the earth is known as incoming solar radiation which in short is termed as *insolation*.

2 On an average the earth receives **1.96 calories per sq.cm per minute or 1.96 Langley(Ly)** at the top of its atmosphere. This is known as **solar constant**.



INSOLATION

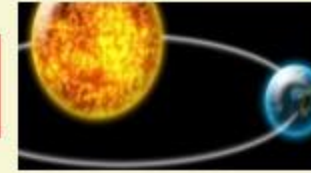


IMPORTANT CONCEPT FOR UPSC PRELIMS MCQs

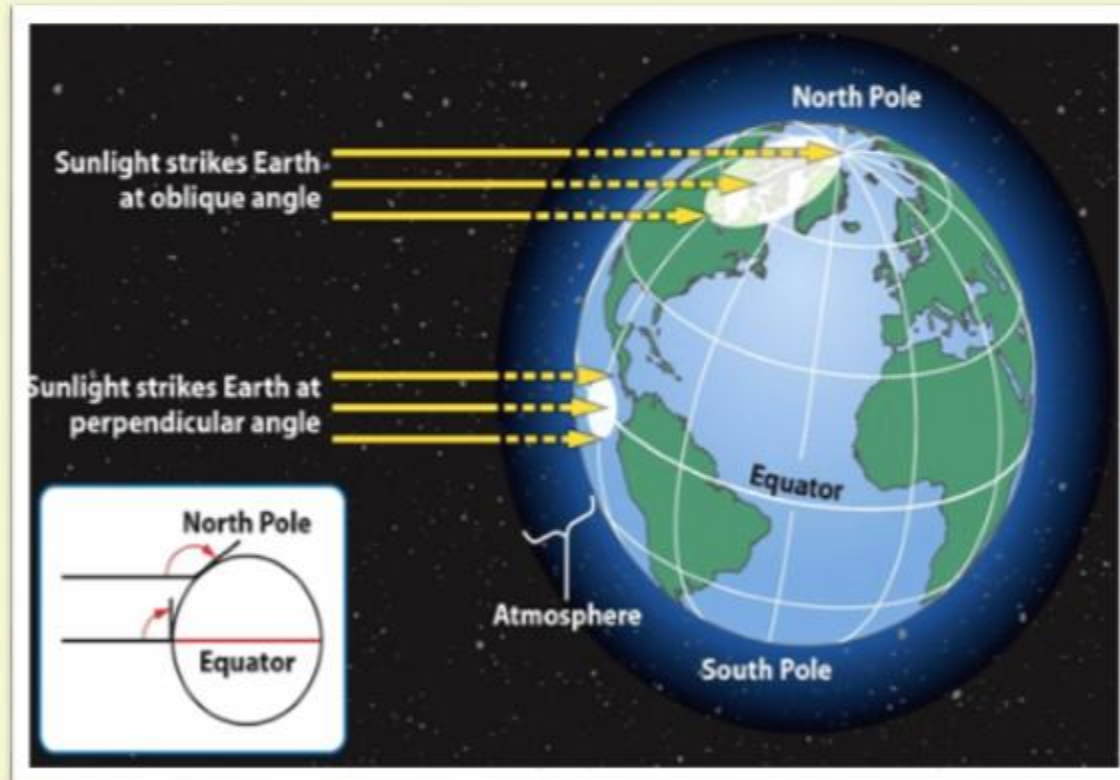
1. Factors controlling DISTRIBUTION OF INSOLATION (covered in this module)
2. Factors controlling DISTRIBUTION OF TEMPERATURE
3. Explicit MCQs on DIFFERENTIAL RATE OF COOLING AND HEATING in land and water.



INSOLATION



VARIABILITY OF INSOLATION ON EARTH SURFACE

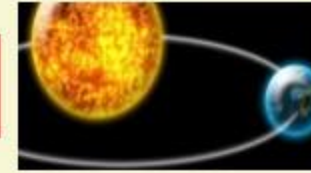


IT DEPENDS ON THE FOLLOWING FACTORS

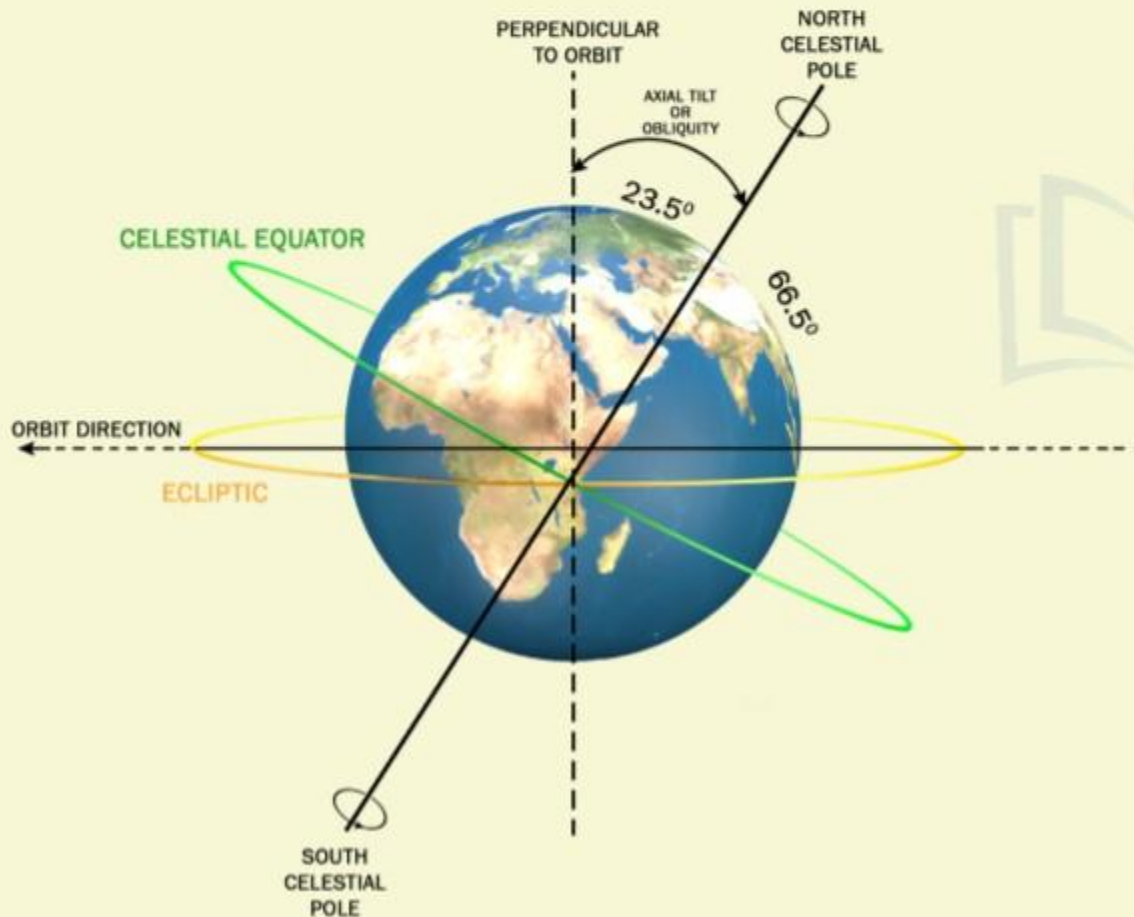
- 1 The angle of the sun's rays;
- 2 The length of the day
- 3 The distance between the Earth and the Sun
- 4 Effects of Atmosphere aka its transparency
- 5 Sun spots



INSOLATION



AXIAL TILT OF THE EARTH

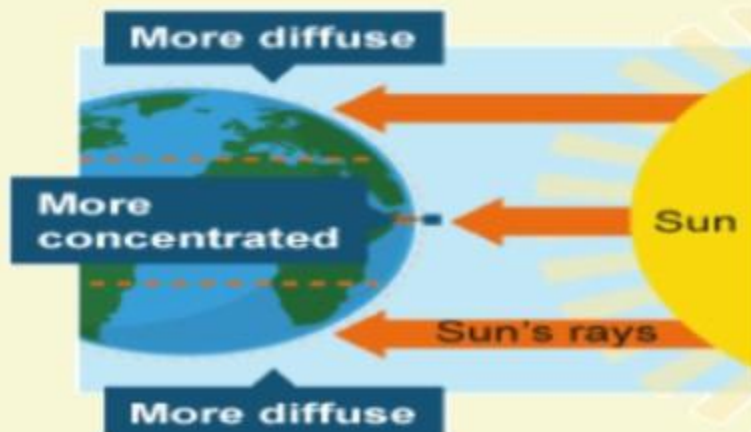
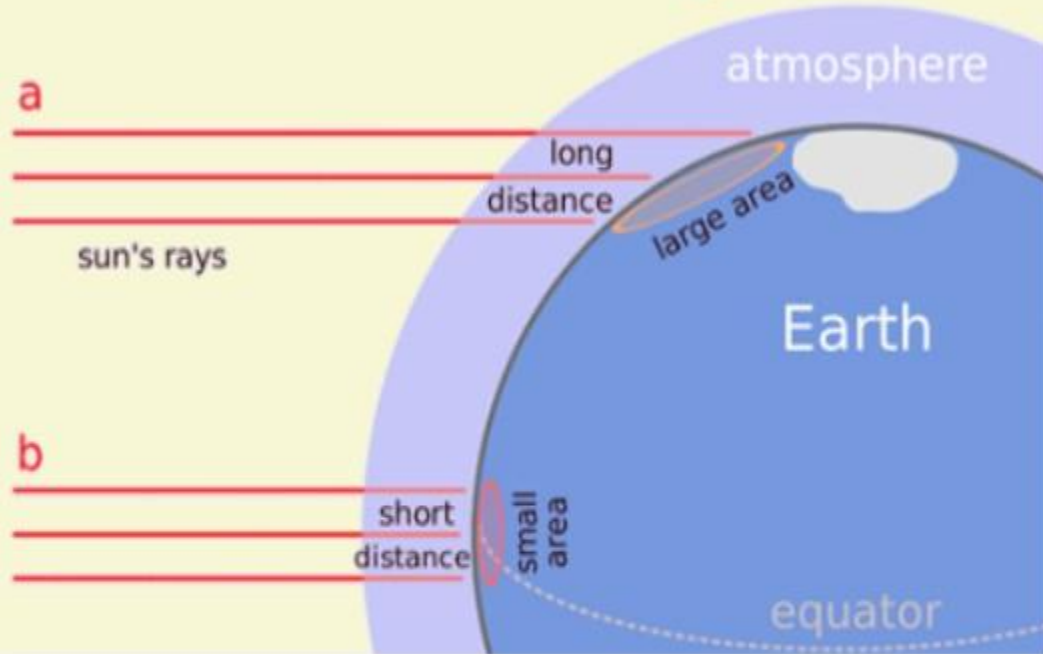
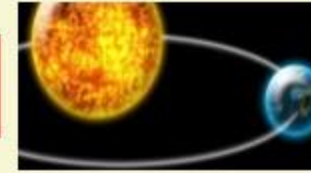


ROTATION OF THE EARTH

- 1 The fact that the earth's axis makes an angle of 66.5° with the plane of its orbit round the sun has a greater influence on the amount of insolation received at different latitudes
- 2 The variations in the duration of the day and angle of sun's rays at different latitudes happen because of this.



INSOLATION



1 ANGLE OF THE SUN'S RAYS

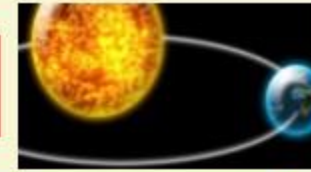
1 The higher the latitude, the lesser is the angle the rays make with the surface of the earth resulting in slant sun rays.

2 The slant rays are required to pass through greater depth of the atmosphere resulting in more absorption, scattering and diffusion.

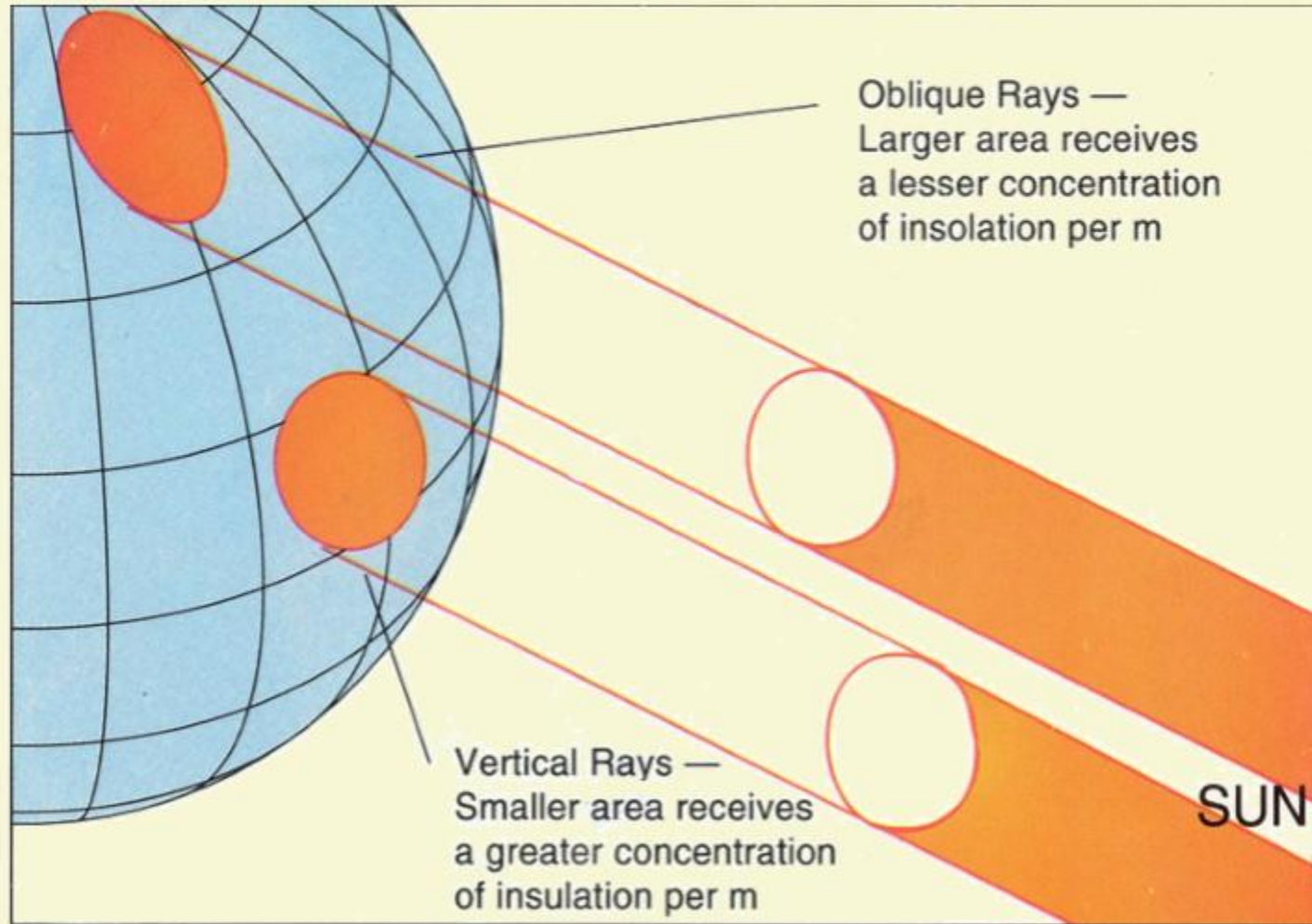
3 The area covered by vertical rays is always less than the slant rays. If more area is covered, the energy gets distributed and the net energy received per unit area decreases.



INSOLATION



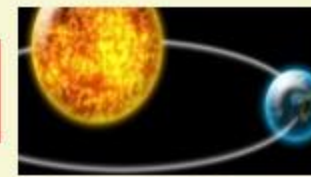
1 ANGLE OF THE SUN'S RAYS



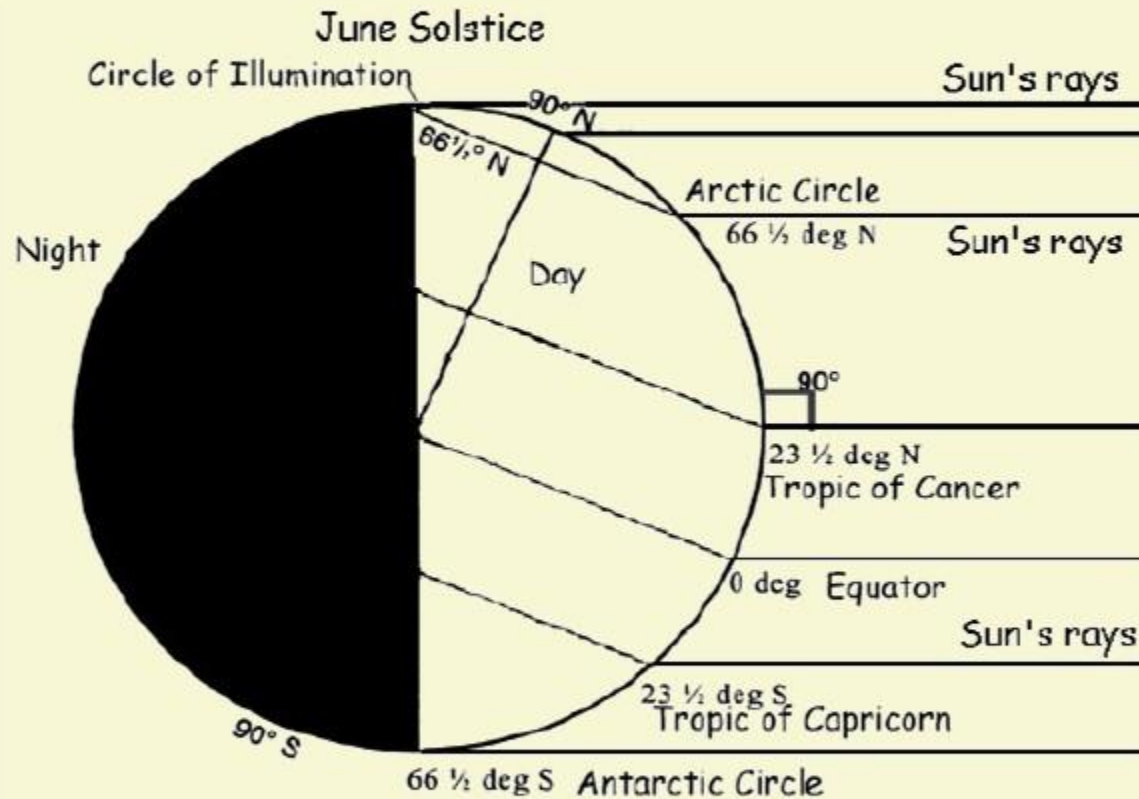
Here the area covered by oblique rays is much more than the area covered by vertical rays.



INSOLATION



2 LENGTH OF THE DAY



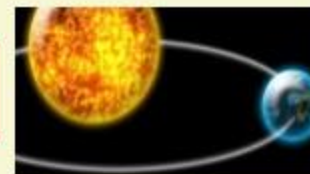
1 If all the other conditions are favourable and equal then longer duration of sunshine and shorter duration of night enable the ground surface to receive larger amount of insolation.

2 The length of day varies at all places except at the equator due to the inclination of the earth's axis, its parallelism and the earth's rotation.

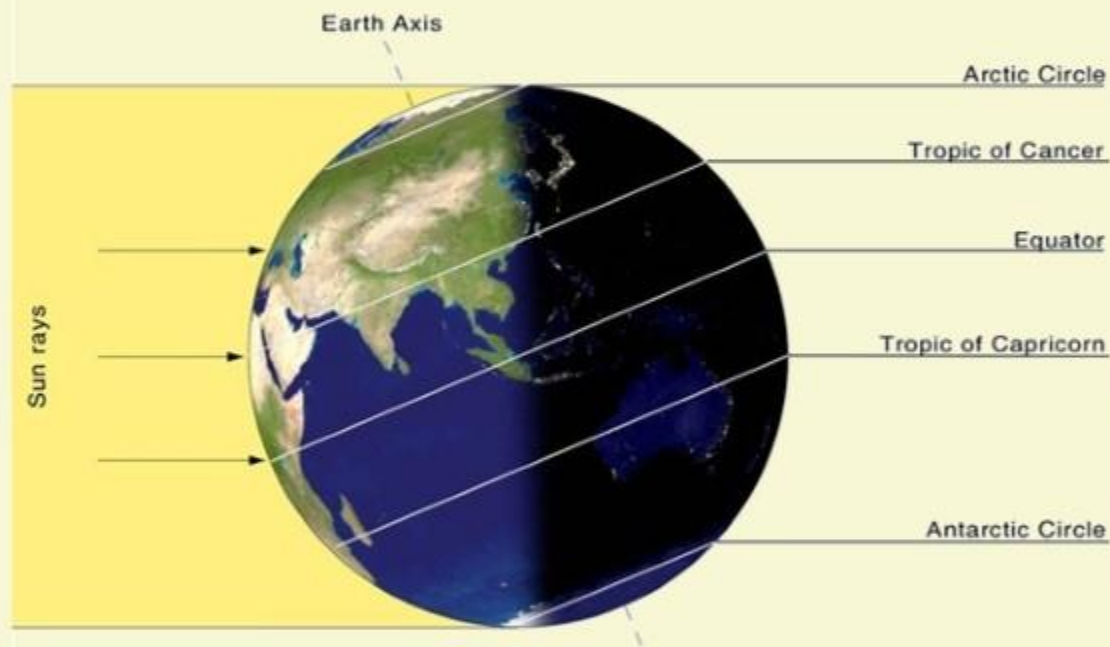
3 The length of day is always of 12 hours at the equator because the circle of illumination always divide the equator into two equal halves.



INSOLATION



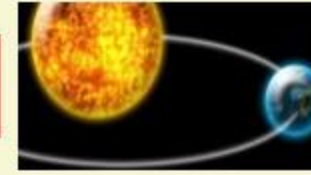
2 LENGTH OF THE DAY



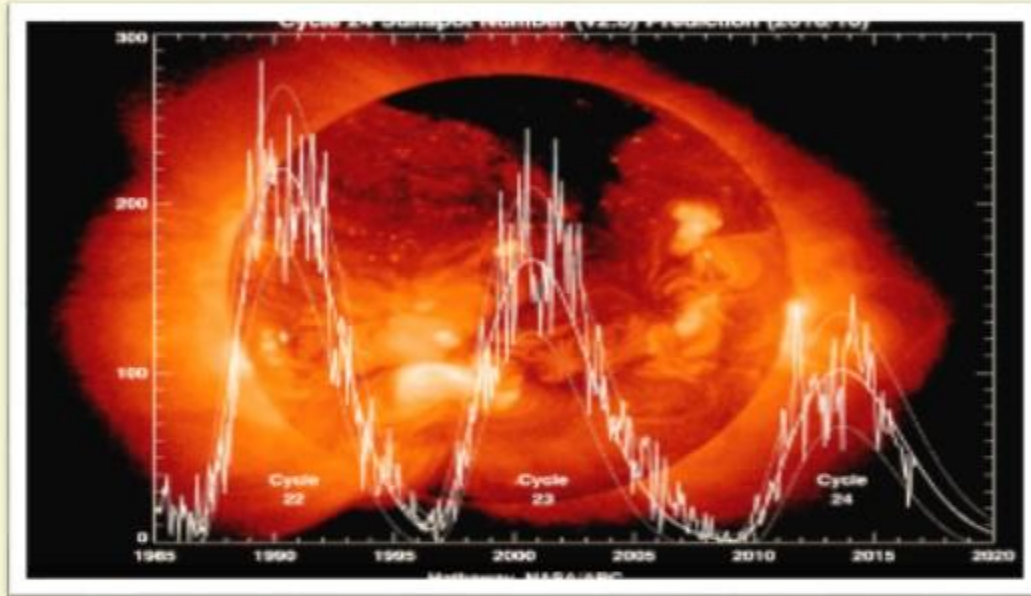
- 1 The fact that the earth's axis makes an angle of 66.5° with the plane of its orbit round the sun has a greater influence on the amount of insolation received at different latitudes.
- 2 The longer the duration of the day, the larger is the amount of insolation.
- 3 But the length of the day increases poleward with the northward march of the sun in the northern hemisphere while it decreases in the southern hemisphere in the mean time.



INSOLATION



3 SUN SPOTS

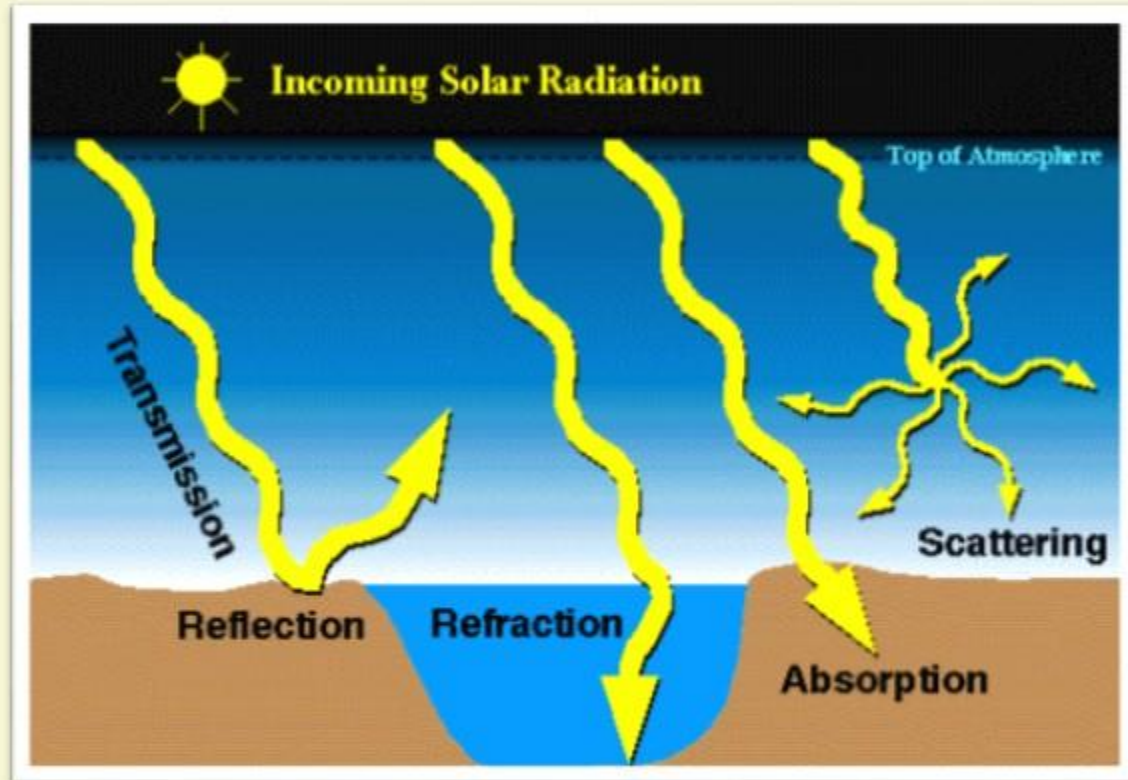
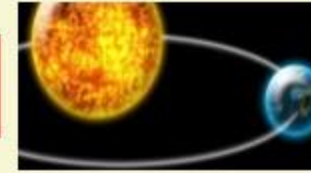


German chemist and amateur astronomer Heinrich Schwabe discovered that there was a fairly regular cycle of change in the number of sunspots and that this cycle lasts about 11 years

- 1 They are magnetic regions created in the solar outer surface due to periodic disturbances and explosions.
- 2 The part of the solar cycle with low sunspot activity is referred to as "solar minimum" while the portion of the cycle with high activity is known as "solar maximum".
- 3 The energy radiated from the sun and therefore the amount of insolation increases when the number of sun spots increases and vice versa.



INSOLATION

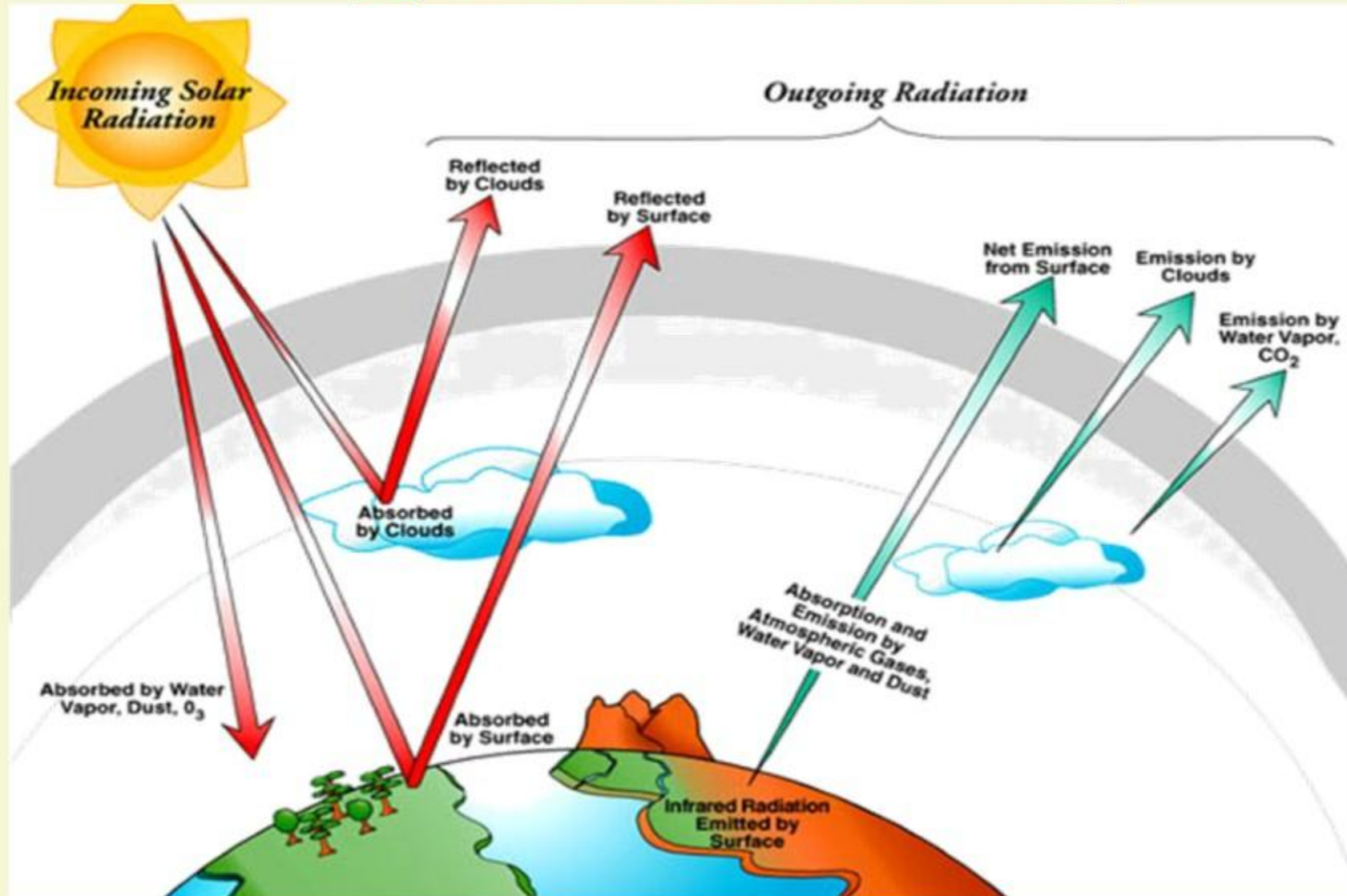


4 EFFECTS OF ATMOSPHERE

- 1 The transparency of the atmosphere is determined by factors like **cloud cover**, **presence of particulate matter**, **mist**, **haze**.
- 2 Solar EM radiation **gets partly absorbed**, **partly reflected**, and **partly scattered**.
- 3 **Thick cloud** hinders the solar radiation to reach the earth's surface. Similarly, **water vapour** absorbs solar radiation resulting in less amount of insolation reaching the surface.

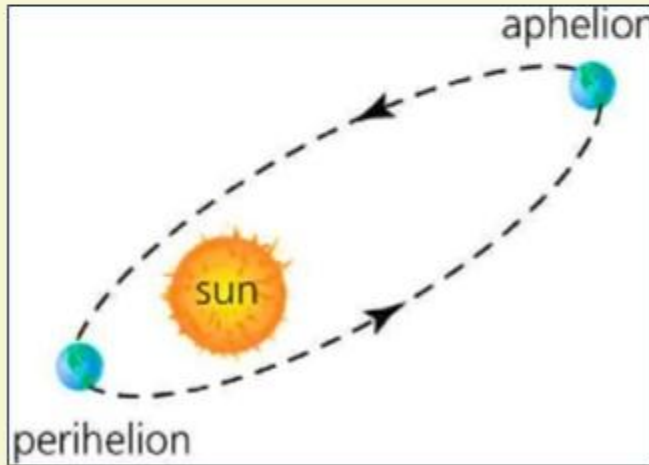
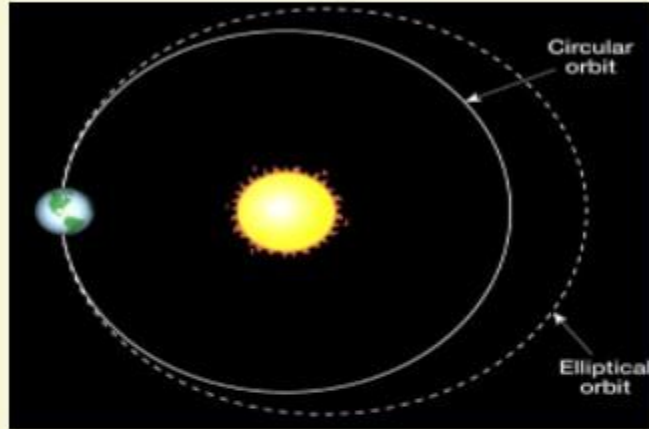
INSOLATION

4 EFFECTS OF ATMOSPHERE



PREVIOUS ITEM DISCUSSED IN MODULE 9 OF GEOMORPHOLOGY

REVOLUTION

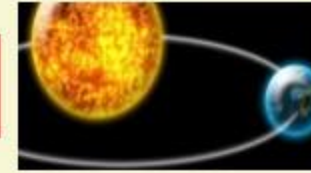


PERIHELION AND APHELION

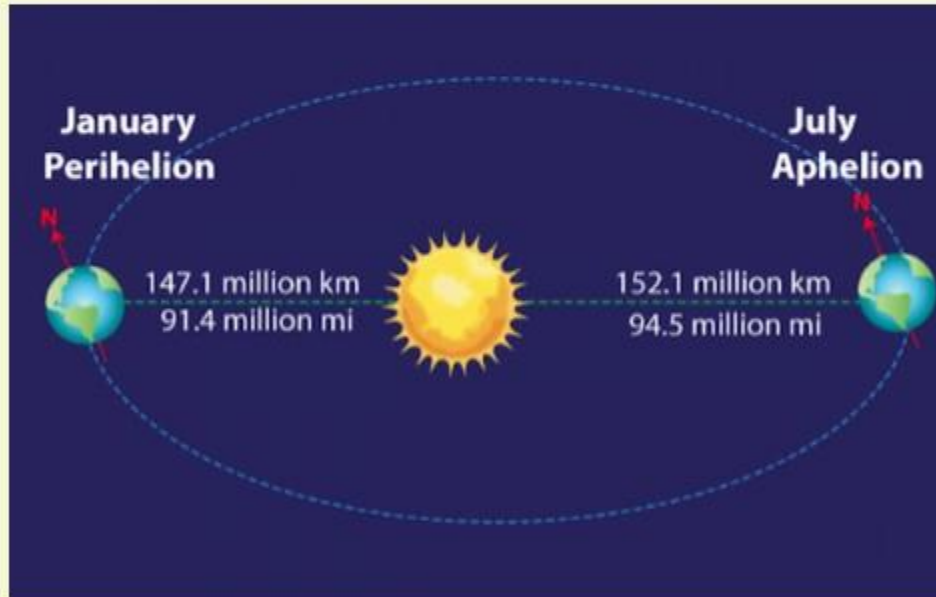
- The Earth is going around the sun in an **elliptical orbit** and throughout its orbit, the Earth is inclined in the same direction.
- The perihelion of any orbit of a celestial body about the Sun is the point where the body comes nearest to the Sun. It is the opposite of aphelion, which is the point in the orbit where the celestial body is farthest from the Sun.



INSOLATION



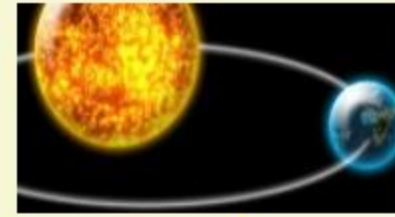
5 DISTANCE BETWEEN THE EARTH AND THE SUN



- 1 As per rule, the Earth at the time of perihelion, when it is nearest to the Sun, should receive maximum insolation while it should receive minimum insolation at the time of aphelion when the Earth is at the greatest distance from the Sun.
- 2 In fact in the month of the January, when the Earth is nearest to the Sun, there is winter season instead of summer season in the northern hemisphere due to low amount of insolation received.
- 3 On the other hand, in the month of July, when the Earth is farthest from the Sun, there is summer instead of winter in the northern hemisphere due to high amount of insolation received.

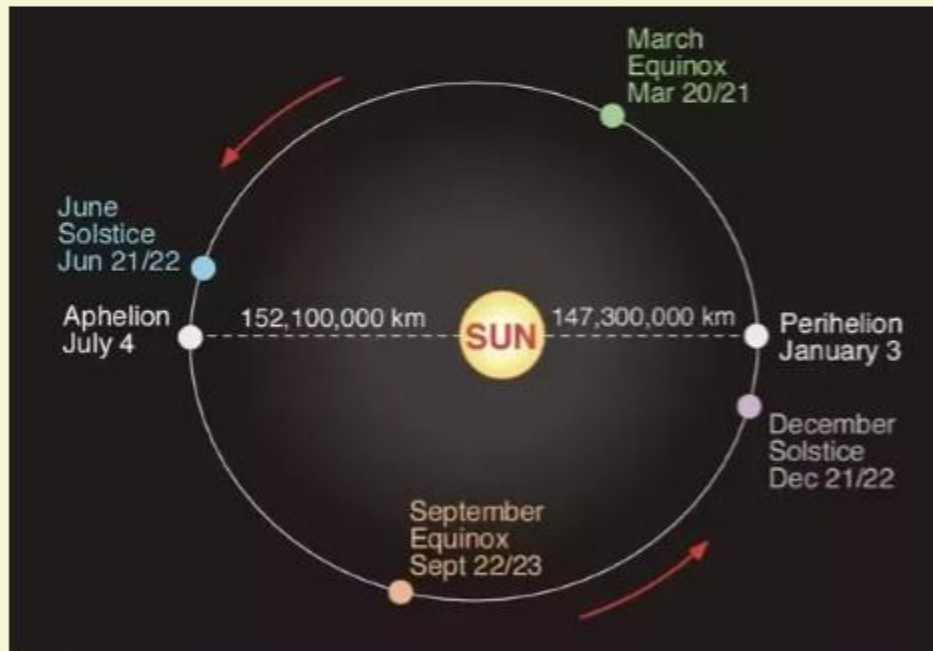


INSOLATION



5 DISTANCE BETWEEN THE EARTH AND THE SUN

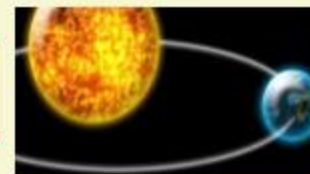
WHAT DOES THIS IMPLY?



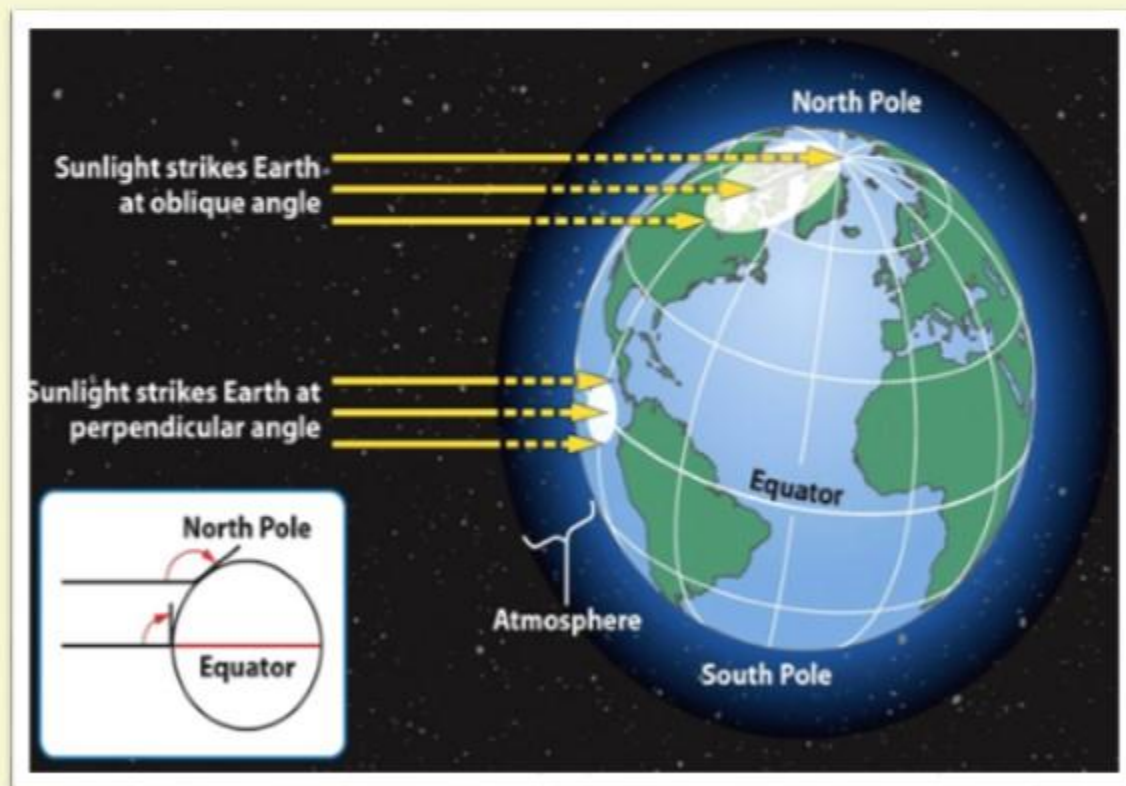
- 4 It is obvious that **factors of the angle of the Sun's rays and the length of the day play more dominant role** in the distribution of insolation than the factor of distance between the Earth and the Sun.
- 5 Thankfully, it is because of this factor **winters are 7% less severe in January in the northern hemisphere but summer is 7% more intense in the southern hemisphere at the time of perihelion.**
- 6 While summer is 7 % less intense in July in the northern hemisphere but winter 7 % is more intense in the southern hemisphere at the time of aphelion.



INSOLATION



VARIABILITY OF INSOLATION ON EARTH SURFACE



IT DEPENDS ON THE FOLLOWING FACTORS

- 1** The angle of the sun's rays;
- 2** The length of the day
- 3** The distance between the Earth and the Sun
- 4** Effects of Atmosphere aka its transparency
- 5** Sun spots

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